

Controlled Evaluation of a ResNet-1D–TCN Pipeline for Single-Channel EEG Sleep Staging: Component Ablations on Sleep-EDF and Zero-Shot Transfer to SHHS1

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Abstract

Background. Comparisons between deep sleep-staging models are frequently confounded, because architecture, preprocessing, subject partitioning and model-selection rules change together across publications. Reported gains therefore cannot be attributed to any single component, and out-of-domain behaviour is rarely characterised beyond a single aggregate score.

Methods. We conducted a pre-specified, subject-wise controlled comparison of six configurations (E0–E4, E6) on the Sleep Cassette subset of Sleep-EDF Expanded (78 subjects, 153 recordings, 195,469 scored 30 s epochs, Fpz–Cz). All configurations shared one 10-fold subject-wise partition, one model-selection rule (validation macro-F1) and one locked test set that was opened exactly once. Configurations differed in a single component at a time, isolating the sequence model (BiLSTM → TCN), the epoch encoder (15-branch CNN → ResNet-1D) and the amplitude-handling scheme. Uncertainty was quantified by paired subject-cluster bootstrap (10,000 resamples) and two-sided Wilcoxon signed-rank tests with Holm correction over four pre-specified comparisons. Generalisation was assessed by locked zero-shot inference on 180 SHHS Visit-1 participants (C4–A1, 169,012 scored epochs); no weights were updated on SHHS.

Results. The ResNet-1D–TCN configuration with fixed-scale amplitude handling (E3) attained the highest out-of-fold macro-F1 (0.7904). Against per-record z -scoring (E6) it gained 0.0213 (CI_{95%} [0.0122, 0.0307]; Holm $p = 0.0012$); a post-protocol repeat with a second seed preserved the direction (+0.0102, CI_{95%} [0.0024, 0.0180]) but not Holm significance. Isolated architectural substitutions produced small, non-significant descriptive gains. E3 was 3.76× faster at inference than the CNN–BiLSTM control, at 4.37× the parameter count and 28.4% higher peak GPU memory. Zero-shot on SHHS1, E3 exceeded the control by 0.0412 subject-mean macro-F1 (CI_{95%} [0.0314, 0.0512]; Holm $p = 2.65 \times 10^{-13}$) and z -scoring by 0.0274. Two novel analyses reframe the findings. First, the entire Sleep-EDF → SHHS1 degradation (Δ macro-F1 = 0.1805) is concentrated in two confusion channels: correcting N3→N2 alone recovers 0.1345 (74.5% of the gap), and correcting N3→N2 together with N2→REM recovers 0.1848, restoring in-domain-level performance. The failure is a slow-wave amplitude-threshold effect (N3 precision 0.944, recall 0.258, falling further to 0.073 at stage boundaries) whose direction is *opposite* to the label-prior shift, which excludes prior shift as its cause and identifies a montage- and age-dependent voltage criterion instead. Second, the null result of the context ablation is explained rather than merely reported: the non-causal TCN already integrates a receptive field of 253 epochs (± 63 min), so the ± 1 -epoch context groups it was asked to supply are nested inside information the sequence model already possesses.

Conclusions. Within this protocol the complete E3 pipeline is the strongest and most transfer-robust configuration, but its advantage is attributable to the evaluated procedure as a whole rather than to architectural optimality. The dominant obstacle to deployment is not aggregate accuracy but a localised, diagnosable and—we argue—correctable montage-dependent failure on slow-wave sleep.

Keywords: sleep staging; single-channel EEG; ResNet-1D; temporal convolutional network; Sleep-EDF Expanded; Sleep Heart Health Study; domain shift; zero-shot evaluation; paired inference.

1. Introduction

Sleep staging assigns one of five labels—W, N1, N2, N3 and REM—to every 30 s epoch of a polysomnographic recording. Manual scoring is slow, requires trained expertise, and retains irreducible disagreement between scorers; in the inter-scorer reliability programme of the American Academy of Sleep Medicine, N1 shows the lowest agreement of all stages [?]. An automated system therefore has to be judged not by overall accuracy alone, which is dominated by the most frequent stages, but by class-balanced metrics, by stability across individuals, and by its behaviour when the recording device, montage or population changes.

Progress in this field is difficult to read from published tables. Studies differ simultaneously in dataset version, subject partitioning, wake-trimming policy, filtering, normalisation, sequence length and model-selection rule, so a reported improvement cannot be attributed to the architecture that the paper nominally proposes. What is missing is not another architecture but a protocol in which one component changes at a time while everything else is held fixed.

We take ZleepAnlystNet [?], a two-stage single-channel model that couples a bank of class-specific convolutional encoders to a BiLSTM, as an internal control, and we substitute two components in isolation: a temporal convolutional network (TCN) as a parallelisable sequence model [?], and a one-dimensional residual network (ResNet-1D) as an epoch encoder [?]. Neither substitution is assumed beneficial. We additionally treat the amplitude-handling scheme as a first-class experimental factor rather than an implementation detail, because it determines whether absolute microvolt relationships survive preprocessing—the quantity most likely to differ between acquisition setups.

The study is organised around four questions. (i) Does a TCN improve on a BiLSTM when the epoch encoder is held fixed? (ii) Does a ResNet-1D encoder help when the sequence model is held fixed? (iii) How does the amplitude-handling scheme affect performance? (iv) Do any of these effects survive zero-shot evaluation on an independent, clinically distinct cohort?

Our contributions are the following.

- (1) A leakage-controlled 10-fold subject-wise protocol in which six configurations share one partition, one selection rule and one test set opened exactly once, with component substitutions designed so that each pre-specified contrast isolates a single change.
- (2) A locked zero-shot transfer campaign to 180 SHHS Visit-1 participants, and a paired, subject-level statistical treatment of it.
- (3) *A decomposition of the domain gap.* We show that the cross-cohort degradation is not diffuse: two confusion channels account for essentially all of it, and the dominant one is a slow-wave amplitude-threshold failure whose signature (very high precision, collapsed recall) and whose direction relative to the label-prior shift jointly identify it as a signal-level, montage-dependent effect rather than a prior-shift or class-imbalance effect.
- (4) *A mechanistic account of a null result.* We show that the context-group ablation could not have detected an effect of the design that was used, because the sequence model’s receptive field already subsumes the context it was being asked to provide—turning an uninformative negative into a constraint on how such ablations must be designed.
- (5) Distribution-free effect sizes and an exact sign test reported alongside every paired comparison, so that the strength of evidence is separable from the sample size.

Throughout, we distinguish pre-specified comparisons, secondary evidence and post-hoc contrasts, and we confine claims to the evidence class that supports them.

2. Related work

DeepSleepNet combined a dual-branch CNN with a BiLSTM to capture epoch morphology and temporal dependence [?]. AttnSleep introduced attention over multi-resolution features [?], and SleepTransformer applied self-attention with explicit uncertainty quantification [?]. Together these establish that sequential context matters substantially. They also illustrate the comparability problem: differences in dataset version, subject split and preprocessing between these works are large enough that cross-table comparison cannot substitute for an ablation carried out inside one protocol, a point emphasised in recent reviews of the field [?].

Sleep-EDF Expanded is the standard public benchmark for single-channel EEG [?, ?]. The Sleep Heart Health Study (SHHS) is a large multi-centre cohort assembled to study sleep-disordered breathing and cardiovascular risk [?], distributed through the National Sleep Research Resource (NSRR) [?]. The two differ in age, comorbidity, recording hardware, montage and label distribution. This makes SHHS a considerably more realistic transfer test than a held-out split of the same corpus, and it is the reason we treat the transfer result—not the in-domain ranking—as the principal evidence about deployability.

Two threads of the literature bear directly on our findings. First, the sleep slow oscillation has a pronounced anterior amplitude maximum and propagates as a travelling wave [?], so the microvolt amplitude of slow-wave activity is derivation-dependent; and slow-wave amplitude declines markedly with age [?]. Both scoring manuals [?, ?] define deep sleep partly through an absolute amplitude criterion, which means an amplitude criterion learned on one derivation and one age band need not transfer. Second, classifier outputs trained under one label prior are miscalibrated under another, and the correction is well characterised [?, ?]. Section ?? uses these two mechanisms as competing explanations and shows that the data discriminate between them.

3. Materials and methods

3.1. Data and labels

The Sleep Cassette subset of Sleep-EDF Expanded comprises 153 recordings from 78 subjects; we use the Fpz–Cz derivation at 100 Hz with 30 s epochs (3,000 samples per epoch). Annotations are mapped to W, N1, N2, N3 and REM; Movement and Unknown epochs retain their temporal position but are excluded from the loss and from all metrics. After windowing, 195,767 epochs remain, of which 195,469 are scored and 298 are Movement/Unknown.

An evaluation window is applied identically to both cohorts. It is anchored on the first and last true sleep epoch (any of N1, N2, N3, REM) and extended by 30 min of wake on each side, i.e. 60 epochs per side. This policy is stated explicitly because it strongly affects the wake fraction and hence every aggregate metric, and because it is one of the principal reasons published numbers are not directly comparable. It is a benchmark window, not a deployable inference rule, since its anchors use ground-truth labels.

The SHHS Visit-1 sample was locked before any prediction was inspected and comprises 180 test participants, with a further 5 adaptation and 15 validation participants that were never used to update weights. The EEG channel is C4–A1 at 125 Hz, resampled to 100 Hz with a polyphase anti-aliasing filter (up 4, down 5, Kaiser window $\beta = 5.0$) applied to the continuous signal before epoching. SHHS labels are mapped $0 \rightarrow W$, $1 \rightarrow N1$, $2 \rightarrow N2$, $\{3, 4\} \rightarrow N3$, $5 \rightarrow REM$, $\{6, 9\} \rightarrow \text{excluded}$. Table ?? reports both label distributions.

The prior shift between cohorts is substantial: total variation distance 0.187, $D_{KL}(EDF \parallel SHHS) = 0.101$ nats. SHHS1 contains proportionally $2.02\times$ as much N3 and $0.38\times$ as much N1. We report the per-class ratios in Table ?? because Section ?? uses their *direction* as evidence.

3.2. Preprocessing conditions

All conditions operate on the continuous calibrated signal in microvolts before epoching. Four conditions enter the comparison:

- **Raw** (E0–E2): no filtering, no clipping, no rescaling; native μV retained.
- **Band-pass only** (E4): zero-phase Butterworth band-pass, order 4, 0.5–30 Hz, implemented as second-order sections with forward–backward filtering.
- **Fixed-scale amplitude handling** (E3): the same band-pass, followed by a hard clip at $\pm 800 \mu V$ and division by a constant factor of 100. The clip threshold is an absolute constant identical for every recording, so no data-dependent statistic is estimated and no leakage is possible; crucially, *relative amplitude between recordings is preserved*.
- **Per-record z-score** (E6): the same band-pass and clip, followed by subtraction of the mean and division by the standard deviation computed over the whole filtered, clipped recording. This is label-independent but *destroys absolute amplitude information across recordings*.

No notch filter is applied: 50 Hz lies outside the 0.5–30 Hz pass-band and at the Nyquist frequency of the 100 Hz signal. The contrast E3 – E6 is thus precisely a contrast between preserving and discarding

Table 1. Scored-label distribution in the two evaluation samples.

Stage	Sleep-EDF		SHHS1 test		
	Epochs	Share	Epochs	Share	Ratio to EDF
W	65,941	33.73%	36,983	21.88%	0.65
N1	21,522	11.01%	7,002	4.14%	0.38
N2	69,132	35.37%	76,287	45.14%	1.28
N3	13,039	6.67%	22,806	13.49%	2.02
REM	25,835	13.22%	25,934	15.34%	1.16
Total scored	195,469	100%	169,012	100%	

Table 2. Configurations compared. Each pre-specified contrast changes exactly one component.

ID	Signal condition	Epoch encoder	Sequence model and role
E0	Raw	15-branch CNN	BiLSTM; internal control
E1	Raw	15-branch CNN	TCN; isolates the sequence model
E2	Raw	ResNet-1D	TCN; isolates the epoch encoder
E3	Band-pass + clip + fixed scale	ResNet-1D	TCN; complete proposed pipeline
E4	Band-pass only	ResNet-1D	TCN; effect of filtering alone
E6	Band-pass + clip + z-score	ResNet-1D	TCN; alternative rescaling

cross-recording amplitude scale, which is why it is the pre-specified comparison of greatest interest for transfer. One further derived variant proved bit-identical to the band-pass-only condition in this corpus and therefore does not constitute an independent experiment.

3.3. Model configurations

Table ?? lists the six configurations.

15-branch CNN encoder. The control encoder instantiates 15 independent small CNNs: three context groups—C (current epoch), P (previous), N (next)—crossed with five class-specific detectors (W, N1, N2, N3, REM). Each branch has 2,495 parameters and comprises two convolutional blocks, Conv1d($1 \rightarrow 10, k=55$) and Conv1d($10 \rightarrow 5, k=25$), each followed by ReLU, batch normalisation and $16 \times$ max-pooling, then a $60 \rightarrow 10 \rightarrow 5$ head. The concatenated output is a 75-dimensional vector per epoch ($15 \text{ branches} \times 5 \text{ softmax outputs}$), ordered C, P, N. Context shifts are clamped at recording boundaries.

ResNet-1D encoder. A stem Conv1d($1 \rightarrow 32, k=50, s=5$) with batch normalisation, ReLU and max-pooling is followed by three residual blocks ($32 \rightarrow 64, s=1$; $64 \rightarrow 128, s=2$; $128 \rightarrow 128, s=2$), each containing two $k=7$ convolutions with a projection shortcut where shape changes. Global average pooling yields a 128-dimensional epoch embedding; total temporal downsampling is $40 \times$. Note that the ResNet-1D encoder consumes *only the current epoch*: it has no C/P/N construction.

Sequence models. The TCN projects the encoder output to 128 dimensions and applies six residual blocks of two $k=3$ convolutions each, with dilations 1, 2, 4, 8, 16, 32, LayerNorm, GELU and dropout 0.2. Padding is symmetric, so the network is *non-causal*, and the receptive field is $1 + 2(k-1) \sum_i d_i = 1 + 4 \times 63 = 253$ epochs, i.e. approximately ± 126 epochs or ± 63 min around the epoch being classified. The BiLSTM control is a single bidirectional layer with 128 hidden units per direction. In both cases the input is the *entire overnight recording* as one sequence (typically 900–1,200 epochs), not a short window; padding is masked at every layer. The receptive-field figure is central to interpreting the context ablation and we return to it in Section ??.

3.4. Two-stage training and subject-wise cross-validation

The 78 subjects are partitioned into 10 subject-disjoint folds. In each evaluation round one fold is the test fold, the next fold (modulo 10) is the validation fold and the remainder are training folds. Both nights of a subject always share the same role. The same partition is used for every configuration, and the test fold never participates in model selection; the training API accepts only training and validation loaders, so test data cannot enter fitting.

Training proceeds in two stages with a *fully frozen* encoder. Stage 1 fits the epoch encoder at epoch level: for E0/E1, the 15 CNN branches are trained independently as five-class problems under class-specific binary weights $w_c = 0.5N/n_c$ and $w_{j \neq c} = 0.5N/(N - n_c)$ computed from training subjects only; for E2–E6, the ResNet-1D and a linear head are trained on single epochs under balanced weights $w_c = N/(5n_c)$, again from training subjects only. The best checkpoint is then verified by SHA-256, switched to evaluation mode, and used under `torch.no_grad()` to extract and cache features. Stage 2 fits the sequence model on those cached features; because the inputs are arrays, the encoder is not in the computation graph and the freeze is absolute.

Optimisation settings are: 15-branch CNN, Adam, lr 10^{-3} , batch 64 epochs, patience 10; ResNet-1D, AdamW, lr 10^{-3} , weight decay 10^{-4} , batch 64, max 40 epochs, patience 8; BiLSTM, Adam, lr 10^{-2} , batch 4 recordings, patience 10; TCN, Adam, lr 5×10^{-4} , batch 8 recordings, max 300 epochs, patience 30. Gradient-norm clipping at 1.0 is applied to sequence models. There is no learning-rate schedule, no label smoothing and no data augmentation. Sequence models use unweighted masked cross-entropy, so their outputs are calibrated to the Sleep-EDF label prior—a fact that Section ?? exploits. Model selection uses validation macro-F1 (validation weighted loss for the CNN branches). E1 reuses the E0 encoder checkpoints unchanged, so E1 – E0 isolates the sequence-model substitution exactly.

3.5. Test locking, statistical analysis and effect sizes

All six configurations completed validation-based selection before the test set was opened, once. In Sleep-EDF, pooled macro-F1 is the descriptive headline metric, but the unit of statistical inference is the subject. Paired subject-cluster bootstrap with 10,000 resamples estimates confidence intervals for the difference in pooled macro-F1, while a two-sided Wilcoxon signed-rank test [?] assesses the subject-level distribution of per-subject differences [?]. Holm correction [?] is applied over the four pre-specified comparisons E1 – E0, E2 – E1, E3 – E2 and E3 – E6. These two procedures target different estimands—a pooled, epoch-weighted quantity and a subject-averaged rank quantity—and may therefore disagree; where they do, we say so explicitly rather than reporting only the favourable one. E3 – E0 is labelled post-hoc throughout.

To make the strength of evidence separable from sample size we additionally report, for every paired comparison, a distribution-free effect size derived from the signs of the paired differences,

$$\Delta_{\text{sign}} = \frac{W - L}{W + L}, \quad (1)$$

where W and L are the numbers of subjects improved and degraded. This is Cliff’s dominance statistic [?] evaluated on the sign of the paired difference; it equals $2\hat{p}_{\text{win}} - 1$ and lies in $[-1, 1]$. We accompany it with the exact two-sided binomial sign test. These statistics use less information than the Wilcoxon test—they ignore magnitudes—so they are reported as unadjusted supplementary robustness statistics, not as members of the pre-specified family. Their value is that they are invariant to the metric’s scale and immune to a few large-magnitude subjects.

After the primary results had been observed, the whole campaign was repeated with a second fixed training seed on the same partition. The two seeds are analysed separately; p -values are never pooled, and two fixed seeds are not treated as a sample from the distribution over initialisations. This repeat is reported as a post-protocol sensitivity analysis.

For SHHS, fold predictions are combined at subject level. The pre-locked primary criterion is subject-mean macro-F1, with E3 – E0 and E3 – E6 as the two locked comparisons under the same bootstrap and Wilcoxon–Holm treatment. The E1/E2 component analysis is secondary evidence on an already-opened sample, and E3 – E2 is explicitly post-hoc.

3.6. Supporting evaluations

Inference timing was measured on identical hardware (Tesla V100) for all configurations and excludes I/O, preprocessing and training. Measurements comprise three rounds of 100 forward passes, and the configuration order was shuffled independently in each round to remove ordering and thermal bias; we report the median across rounds. Feature spaces were compared by silhouette coefficient [?] on a stratified subsample of 200 epochs per class per fold (1,000 epochs per fold, 10,000 in total) after reduction to 20 principal components; this subsampling is a limitation of that analysis and we treat it as descriptive only. The context-group ablation retains the sequence model unchanged and replaces the ablated group with its per-dimension mean estimated on valid training epochs, preserving input dimensionality; its primary criterion is macro-F1 restricted to a transition neighbourhood of radius $r = 1$ epoch, with $r = 2$ as a sensitivity check, and 10,000 bootstrap resamples.

4. Results

4.1. Out-of-fold performance on Sleep-EDF

E3 is descriptively highest on all four metrics (Table ??). A pooled ranking is not by itself statistical evidence, and Table ?? shows how little of this ordering survives inference.

Figure ?? shows these effects graphically. Only E3–E6 is consistent across all three lenses: a fully positive bootstrap interval, Holm significance, and a sign test that agrees. The remaining pre-specified contrasts have intervals containing zero and do not reach significance.

E3–E2 deserves particular attention because it is the clearest case of estimand disagreement in the study. Its pooled bootstrap interval marginally excludes zero ($[0.000305, 0.014520]$), yet only 37 of 78 subjects improve, $\Delta_{\text{sign}} = -0.051$, and both the Wilcoxon and sign tests are far from significance ($p_{\text{sign}} = 0.734$). Pooled macro-F1 is epoch-weighted and is therefore dominated by subjects contributing more scored epochs; the subject-level statistics weight every subject equally. When these disagree, the honest reading is that the effect is carried by a subset of high-epoch-count subjects and is not a general property of the population. We do not claim E3 improves on E2 in-domain.

For the post-hoc E3–E0 contrast, the added macro-F1 of the complete pipeline is 0.0150 with a fully positive interval; because it changes architecture and preprocessing simultaneously, it supports the pipeline as a whole and cannot attribute credit to any component.

4.2. Per-class error structure on Sleep-EDF

Table ?? breaks performance down by stage. N1 is by far E3’s hardest class, which is consistent with it also being the stage of lowest human inter-scorer agreement [?]. The largest advantage of E3 over E6 is on N3 (+0.0628), and the gain is driven mainly by precision (0.807 vs 0.716). This is the first indication that the amplitude-handling scheme acts primarily through slow-wave sleep, and it anticipates the transfer analysis in Section ??.

A useful summary of in-domain calibration is the ratio of predicted to true epochs per class. For E3 on Sleep-EDF these ratios are W 0.993, N1 1.051, N2 0.984, N3 1.003, REM 1.017: the model reproduces the marginal label distribution almost exactly. This near-perfect in-domain marginal calibration is the baseline against which the SHHS behaviour must be read.

In the pooled confusion matrix (Table ??), the dominant error channels are the reciprocal $N2 \leftrightarrow N1$ confusions (5,496 and 4,867) and $W \rightarrow N1$ (3,888), i.e. the in-domain error budget is concentrated on N1 boundaries—the expected behaviour for a well-fitted single-channel model.

4.3. Sensitivity to the training seed

Table ?? lists the per-configuration figures. All four pre-specified differences keep a positive sign in both seeds, and E3–E6 is the only contrast whose bootstrap interval is fully positive in both runs. However, its Holm-adjusted significance appears in seed 42 only, and its magnitude halves ($0.0213 \rightarrow 0.0102$). The defensible conclusion is that the *direction* of the amplitude-handling effect is stable while its *inferential strength* is not. Note also that E6 is the configuration most sensitive to the seed (+0.0089), which is itself

Table 3. Out-of-fold test performance on Sleep-EDF (seed 42, pooled over all folds).

Configuration	macro-F1	Accuracy	κ	F1 (N1)
E0	0.7754	0.8262	0.7614	0.5009
E1	0.7802	0.8315	0.7683	0.5115
E2	0.7835	0.8341	0.7716	0.5180
E3	0.7904	0.8363	0.7754	0.5276
E4	0.7891	0.8360	0.7746	0.5270
E6	0.7691	0.8230	0.7573	0.5124

Table 4. Pre-specified paired comparisons on Sleep-EDF macro-F1 ($n = 78$ subjects). Holm correction is over the four pre-specified rows; sign-test statistics are unadjusted supplementary robustness checks. E3 – E0 is post-hoc and is excluded from the Holm family.

Comparison	Δ macro-F1	CI _{95%}	p_{Holm}	W/T/L	Δ_{sign}	p_{sign}
E1 – E0	0.004811	[−0.000915, 0.010193]	0.1022	51/0/27	+0.308	0.0088
E2 – E1	0.003251	[−0.002370, 0.008808]	0.1036	44/0/34	+0.128	0.3082
E3 – E2	0.006962	[0.000305, 0.014520]	0.8989	37/0/41	−0.051	0.7343
E3 – E6	0.021319	[0.012179, 0.030698]	0.0012	49/0/29	+0.256	0.0308
<i>E3 – E0 (post-hoc)</i>	0.015024	[0.005746, 0.025871]	—	49/0/29	+0.256	0.0308

consistent with per-record z -scoring producing a less well-conditioned learning problem. Two fixed seeds cannot characterise the distribution over initialisations, and we make no claim of replication.

4.4. Complexity, latency and training cost

As Table ?? shows, the ResNet-1D-TCN family is $3.76\times$ faster than the BiLSTM control at inference while carrying $4.37\times$ the parameters and 28.4% more peak memory. The advantage (Figure ??) is therefore one of *parallelism*, not of model compactness: the TCN evaluates all epochs of a recording concurrently whereas the BiLSTM must unroll sequentially. Notably, E1 – E0 gives only a $1.07\times$ speed-up, which shows that the speed benefit arises from the ResNet-1D encoder replacing 15 sequentially-scheduled small CNN branches, not from the TCN alone. We state this because it corrects a natural but wrong inference from the headline number.

End-to-end (Table ??), E2 trained $12.2\times$ faster than E0 and E3 $10.2\times$ faster. E1’s 14 min total is not comparable to the others, because E1 reuses E0’s encoder checkpoints and therefore pays only the sequence-model stage; the figure is nevertheless informative, since it isolates that stage as a small fraction of total cost and locates the expense in encoder training. These are wall-clock observations under one scheduling regime, early-stopping configuration and hardware, and should not be read as hardware-independent speed claims.

4.5. Feature geometry

Across all ten folds without exception, the ResNet-1D embedding has a lower silhouette coefficient than the 15-branch CNN feature vector: mean 0.1048 (SD 0.0229) versus 0.2126 (SD 0.0356), mean difference -0.1079 (SD 0.0228; range -0.1526 to -0.0608). The consistency is total (10/10 folds; Figure ??a), so the effect is not noise.

The correct reading is not that the ResNet embedding is worse. Silhouette rewards compact, Euclidean, per-epoch clusters, whereas the downstream TCN integrates a ± 126 -epoch neighbourhood and can exploit distributional or temporal structure that no per-epoch clustering index measures. The 15-branch CNN’s features are, by construction, 75 calibrated class posteriors—a representation almost engineered to score well on silhouette—while the ResNet’s 128 dimensions are unconstrained. That E2 nonetheless matches or

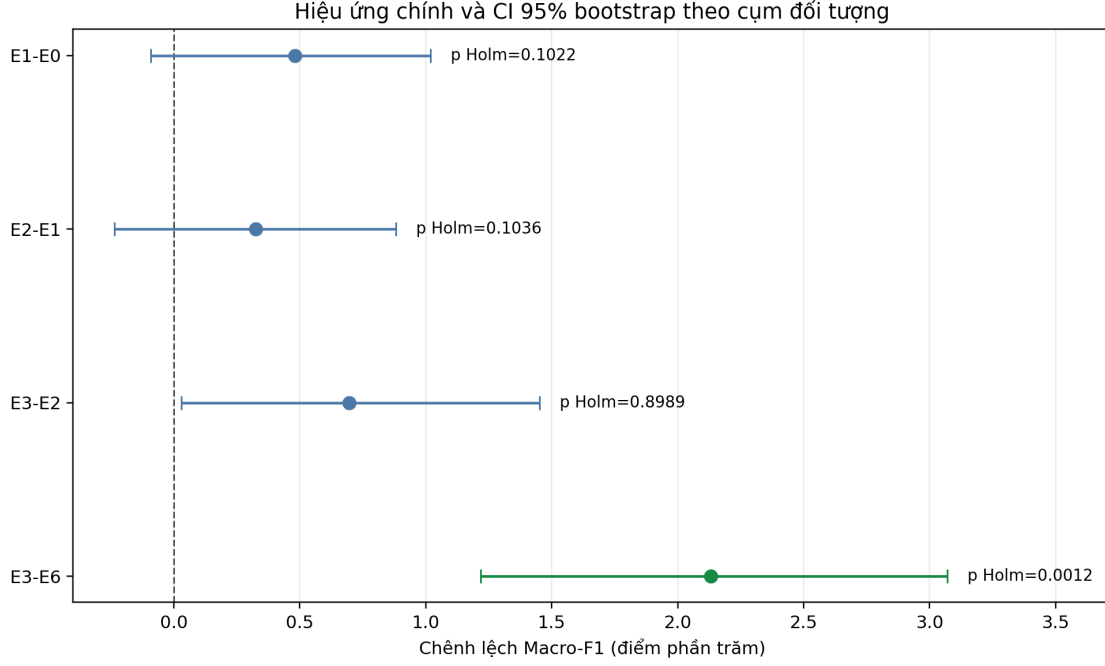


Figure 1. Paired macro-F1 differences with subject-cluster bootstrap intervals on Sleep-EDF.

Table 5. Per-class metrics on Sleep-EDF: E3 versus E6 (pooled out-of-fold predictions, seed 42). Descriptive statistics on the same test predictions; not a new test family.

Stage	E3 Prec.	E3 Rec.	E3 F1	E6 Prec.	E6 Rec.	E6 F1	$\Delta F1$
W	0.9338	0.9271	0.9304	0.9391	0.9169	0.9279	+0.0026
N1	0.5149	0.5410	0.5276	0.5032	0.5219	0.5124	+0.0152
N2	0.8583	0.8447	0.8514	0.8401	0.8368	0.8385	+0.0130
N3	0.8072	0.8097	0.8085	0.7160	0.7779	0.7457	+0.0628
REM	0.8274	0.8413	0.8343	0.8227	0.8197	0.8212	+0.0131

exceeds E1 downstream (Table ??) is direct evidence that per-epoch cluster separability is a poor proxy for sequence-level utility, and we report the silhouette result mainly to document that negative finding.

4.6. Context-group ablation

No comparison approaches significance; every interval comfortably contains zero and all Holm-adjusted p -values equal 1.000. On the primary criterion the full context condition actually loses on a majority of subjects (36/78 wins). Within this design there is no detectable added value of the neighbouring context groups at stage transitions. Section ?? argues that this outcome was structurally predetermined and explains what it does and does not license.

Three descriptive observations deserve recording, with the explicit caveat that none is statistically supported.

First, the effect of P and N is *asymmetric*: adding the next epoch to the current one raises overall macro-F1 by 0.0041, whereas adding the previous epoch raises it by 0.0003—an order of magnitude less. CN is in fact the descriptively best condition overall, exceeding the full CPN condition, and FULL – CN is negative on overall macro-F1 (–0.0014) and on accuracy (–0.0011).

Second, the two groups are *not additive*. The interaction index FULL – CP – CN + C is –0.00161 over the

Table 6. Pooled confusion matrix of E3 on Sleep-EDF (rows: reference; columns: prediction).

Reference \ Predicted	W	N1	N2	N3	REM
W	61,133	3,888	416	56	448
N1	3,283	11,643	4,867	105	1,624
N2	539	5,496	58,394	2,271	2,432
N3	30	45	2,377	10,558	29
REM	483	1,542	1,984	90	21,736

Table 7. Out-of-fold test macro-F1 for two fixed seeds. Seed 42 is the primary campaign; seed 123 is a post-protocol repeat on the same partition.

Configuration	Seed 42	Seed 123	Δ
E0	0.7754	0.7755	+0.0000
E1	0.7802	0.7776	−0.0026
E2	0.7835	0.7828	−0.0007
E3	0.7904	0.7883	−0.0022
E4	0.7891	0.7887	−0.0004
E6	0.7691	0.7780	+0.0089

whole corpus and +0.00258 in the transition region, i.e. it changes sign between the two criteria. Whatever P and N contribute, they do not contribute independently, and no single scalar summarises their joint effect. This is the quantitative reason we regard any statement of the form “group X carries $k\%$ of the information” as unsupportable here, and we recommend that the earlier gloss of this kind be withdrawn.

Third, the transition-pair breakdown is not uniform (Table ??): relative to C alone, full context helps at $W \leftrightarrow N1$ and $N1 \leftrightarrow REM$ but hurts at $N1 \leftrightarrow N2$ and $N2 \leftrightarrow N3$, and the best condition differs by pair. A forward asymmetry would be physiologically coherent, since several scoring rules are prospectively defined—N2 onset is fixed by the first K-complex or spindle, and REM continuation rules look ahead [?][—]but all these differences are far smaller than their confidence intervals and we advance them only as hypotheses for a study designed to test them.

4.7. Zero-shot evaluation on SHHS1

This replication (Table ??) is an important asymmetry in the evidence base. In domain, the E3 – E6 effect lost Holm significance under the second seed (Table ??); out of domain, both locked contrasts retain significance and the configuration ordering is unchanged. Transfer performance is therefore the more stable of the two readouts in this study, which is the opposite of the usual expectation and argues for treating the out-of-domain campaign, not the in-domain one, as the primary evidence about the pipeline.

Both locked comparisons (Table ??) are unambiguous, and the sign-based effect sizes are large ($\Delta_{\text{sign}} = +0.533$ and $+0.389$; exact sign tests $p = 3.8 \times 10^{-13}$ and 1.9×10^{-7}). Supporting metrics (Figure ??) move together: relative to E0, E3 gains 0.0338 pooled macro-F1, 0.0368 accuracy and 0.0461 κ . Within ± 1 epoch of a transition, E3 gains 0.0366 over E0 and 0.0183 over E6.

The component analysis is more equivocal. E1 – E0 remains positive but small, supporting the TCN substitution. E2 – E1 is *negative* out of domain (−0.0128, interval entirely below zero), directly contradicting the in-domain ordering: the ResNet-1D encoder does not generalise better than the 15-branch CNN encoder on unprocessed SHHS signals. The post-hoc E3 – E2 contrast is by far the largest effect in the entire study (+0.0475, $\Delta_{\text{sign}} = +0.633$, 147/180 subjects improved). Since E3 and E2 share an identical architecture and differ only in preprocessing, this isolates preprocessing—not architecture—as the dominant determinant of transfer in this study.

A seed-123 extension evaluating E4 (band-pass only) reinforces this. E4 reached subject-mean macro-F1 0.5732, pooled macro-F1 0.6147, accuracy 0.7064, $\kappa = 0.5869$ and N1 F1 0.3492, exceeding same-seed

Table 8. Seed sensitivity of the four pre-specified contrasts. Each p -value is Holm-corrected *within* its own seed; no test pools the two seeds.

Comparison	Seed 42: Δ [CI], p_{Holm}	Seed 123: Δ [CI], p_{Holm}
E1 – E0	0.004811 [−0.000915, 0.010193], 0.1022	0.002188 [−0.002870, 0.007043], 0.9130
E2 – E1	0.003251 [−0.002370, 0.008808], 0.1036	0.005143 [−0.000571, 0.011292], 0.2400
E3 – E2	0.006962 [0.000305, 0.014520], 0.8989	0.005478 [−0.002611, 0.015796], 0.9130
E3 – E6	0.021319 [0.012179, 0.030698], 0.0012	0.010249 [0.002435, 0.017989], 0.1313

Table 9. Forward-inference profile on a Tesla V100 (fold 0, seed 42). Throughput in epochs s^{-1} ; peak memory in MiB.

ID	Parameters	Median (ms)	Throughput	Peak mem.	Speed-up
E0	248,630	13.4315	7,445	58.81	1.00×
E1	640,950	12.5111	7,993	60.31	1.07×
E2	1,085,578	3.5750	27,972	75.54	3.76×
E3	1,085,578	3.5687	28,021	75.54	3.76×
E6	1,085,578	3.5756	27,967	75.54	3.76×

E2 by 0.0323 and same-seed E3 by 0.0098. Because E3 and E4 differ only by clipping and a constant rescaling, and both preserve cross-recording amplitude relationships, this small difference does not overturn the amplitude account developed below; it does indicate that the clip-and-rescale step carries no transfer benefit over band-pass filtering alone, and that band-pass filtering is doing most of the work. These are paired comparisons on an already-opened sample; they are neither equivalence nor non-inferiority tests.

4.8. Where the domain gap actually lives

The aggregate transfer statistics above are conventional; this section is not. E3’s macro-F1 falls from 0.7904 in domain to 0.6099 out of domain, a gap of 0.1805. We show that this gap is not a diffuse degradation but is almost entirely localised in two confusion channels, and that its mechanism can be identified from the data already collected.

Three facts in Table ?? constrain the explanation.

(i) The degradation is dominated by one class. N3’s F1 falls by 0.4030; the other four classes fall by 0.09–0.18. N3 alone contributes 45% of the total macro-averaged drop.

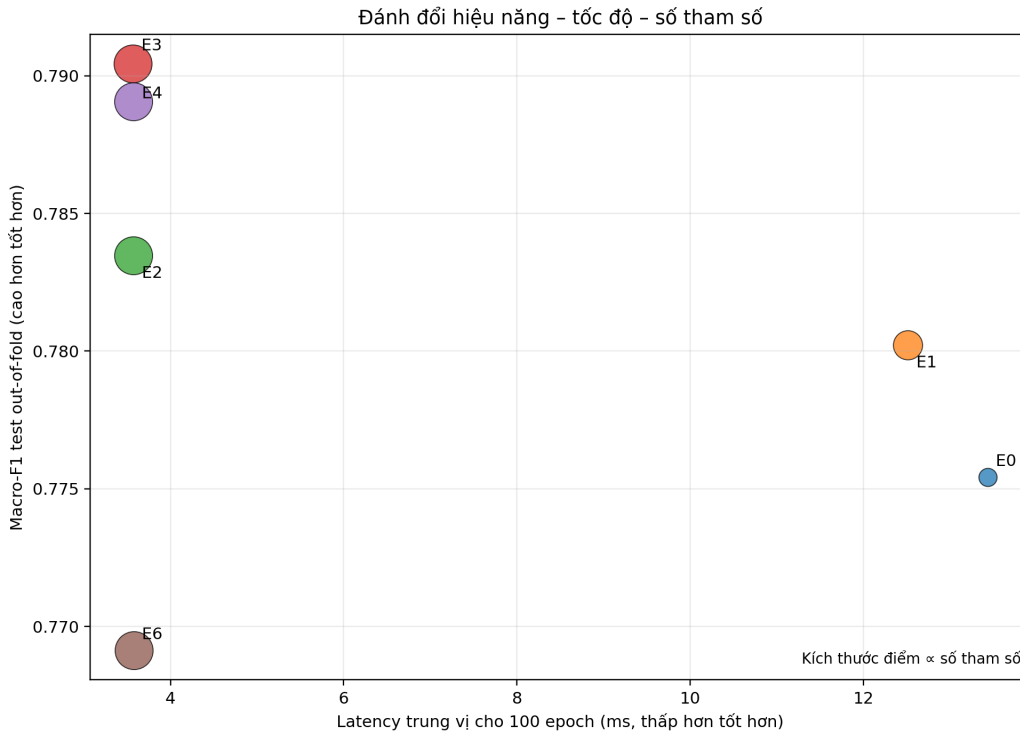
(ii) The N3 failure has a threshold signature. N3 precision is *higher* out of domain than in (0.944 vs 0.807) while recall collapses (0.258 vs 0.810). The model emits N3 only 0.273× as often as it should, and 16,674 of 22,806 true N3 epochs—73.1%—are labelled N2. A model that is nearly always right when it fires but fires far too rarely is behaving like a detector whose decision threshold sits too high for the incoming signal. It is not confused about what N3 looks like; it is not seeing enough of it.

(iii) The miscalibration runs opposite to the label-prior shift. This is the decisive observation. SHHS1 contains 2.02× as much N3 as Sleep-EDF, yet the model predicts N3 at 0.273× the required rate. It contains 0.38× as much N1, yet the model over-predicts N1 by 2.151×. If the failure were driven by prior shift—the standard explanation for transfer degradation in an unweighted classifier—the errors would run *with* the prior change, and the standard correction [?] would repair them. Here that correction would make N3 strictly worse. Prior shift is therefore excluded as the primary cause, and the cause must lie at the signal level.

Table ?? makes the localisation quantitative. Repairing the single N3→N2 channel recovers 74.5% of the entire cross-cohort gap. Repairing N3→N2 together with N2→REM yields macro-F1 = 0.7947, marginally *above* the in-domain figure of 0.7904. In other words, outside these two channels the model transfers to a

Table 10. Wall-clock training and validation cost of the seed-123 campaign (10 folds, sequential, same GPU). The test set was not opened during these runs.

Configuration	Total (10 folds)	Mean per fold
E0 — 15-branch CNN + BiLSTM	33 h 35 min 36 s	3 h 21 min 34 s
E1 — 15-branch CNN + TCN	14 min 17 s	1 min 26 s
E2 — ResNet-1D + TCN	2 h 44 min 57 s	16 min 30 s
E3 — ResNet-1D + TCN, fixed scale	3 h 16 min 40 s	19 min 40 s
E4 — ResNet-1D + TCN, band-pass	2 h 55 min 56 s	17 min 36 s
E6 — ResNet-1D + TCN, z-score	2 h 24 min 44 s	14 min 28 s
All six configurations	45 h 12 min 10 s	4 h 31 min 13 s

**Figure 2.** Trade-off between macro-F1, latency and parameter count. Marker area scales with parameters.

clinically distinct, older, differently-instrumented cohort with essentially no loss. This is a very different—and far more actionable—picture than "performance degrades by 0.18 under domain shift".

The remaining channel of Table ??, N2→REM (11,605 epochs, 15.2% of true N2), shows the mirror-image signature: REM is over-emitted at $1.477\times$ with recall 0.894 but precision only 0.605. Without EOG or EMG, REM must be inferred from low-amplitude mixed-frequency EEG, and in older adults N2 with reduced spindle density presents a similar single-channel picture. E2 is worse still on this channel (REM precision 0.489), so preprocessing mitigates but does not remove it.

For completeness, the per-class comparison against E2 confirms that the E3 preprocessing improves every class out of domain, most in REM (+0.0776) and N1 (+0.0618), yet leaves N3 recall almost untouched (0.2582 vs 0.2496)—the one place where preprocessing does *not* help (Table ??). Any explanation of the N3 collapse must therefore be something that both E2 and E3 share.

Restricting the analysis to the transition neighbourhood sharpens the picture considerably and provides the strongest single piece of evidence for the account developed in Section 5.3. Within ± 1 epoch of a stage change (44,744 epochs, 26.5% of the sample), E3 attains macro-F1 0.4689, accuracy 0.5290 and $\kappa = 0.3642$,

Table 11. Context-group ablation within E1 (15-branch CNN + TCN). C = current epoch, P = previous, N = next; the ablated group is replaced by its per-dimension training mean. Transition macro-F1 is restricted to epochs within $r = 1$ of a stage change (43,237 epochs, 22.1% of the corpus).

Condition	Overall macro-F1	Δ vs C	Transition macro-F1	Δ vs C	F1 (N1)
C (current only)	0.777477	—	0.619102	—	0.5011
CP	0.777733	+0.000257	0.617686	−0.001416	0.5022
CN	0.781585	+0.004108	0.618895	−0.000207	0.5055
FULL (CPN)	0.780230	+0.002753	0.620055	+0.000953	0.5115

Table 12. Paired comparisons for the context ablation (primary criterion: transition-region macro-F1, $r = 1$; 10,000 bootstrap resamples; $n = 78$).

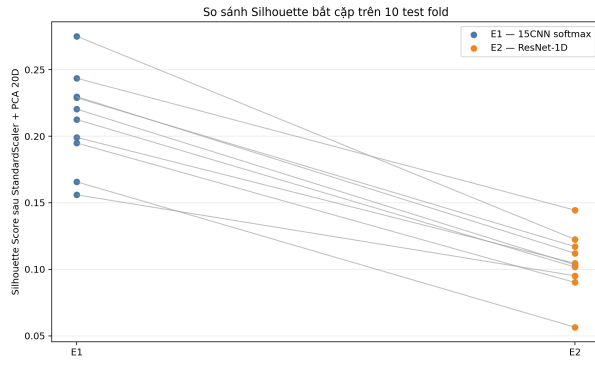
Comparison	Role	Δ transition macro-F1	CI _{95%}	p_{Holm}	W/T/L
FULL – C	primary	0.000953	[−0.004588, 0.006568]	1.000	36/0/42
FULL – CP	secondary	0.002369	[−0.002990, 0.007714]	1.000	41/0/37
FULL – CN	secondary	0.001160	[−0.003918, 0.006490]	1.000	38/0/40

against 0.4219, 0.4910 and 0.3188 for E2 (Table ??). N3 recall there falls to **0.0733** while N3 precision remains 0.831: at stage boundaries the model recovers barely one in fourteen true N3 epochs, yet is right five times out of six when it does commit. Of the 7,563 true N3 epochs in this region, 6,855 (90.6%) are labelled N2. A threshold-based failure should worsen exactly here, because slow-wave activity at the edges of an N3 bout is by definition less established and therefore closest to the amplitude criterion; a failure of representation or of stage concept would not show this gradient.

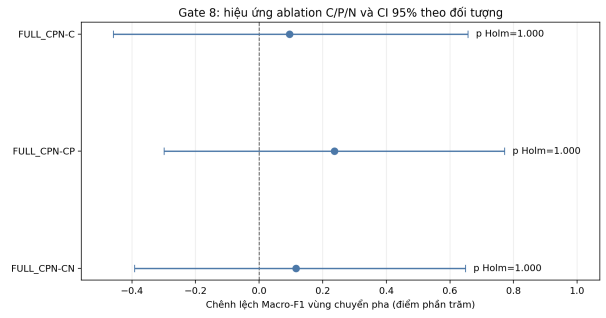
The E3 preprocessing does deliver a large gain on N1 in this region—recall rises by 0.1909 ([0.1513, 0.2297]) over E2 across the whole sample, the single largest per-class effect of preprocessing anywhere in the study—which confirms that the transfer improvement E3 provides is real and is concentrated on the stage that most needs it. It simply does not touch N3.

Table 13. Context ablation broken down by pre-specified transition pair (macro-F1 within $r = 1$). Best condition per row in bold. Differences are descriptive only; no comparison in Table ?? reached significance.

Transition pair	Epochs	C	CP	CN	FULL
W↔N1	12,360	0.3279	0.3330	0.3251	0.3303
N1↔N2	13,057	0.3489	0.3475	0.3440	0.3389
N1↔REM	3,261	0.2651	0.2589	0.2534	0.2690
N2↔N3	10,026	0.4281	0.4268	0.4216	0.4272
All transitions ($r=1$)	43,237	0.6191	0.6177	0.6189	0.6201
All transitions ($r=2$)	58,545	0.6474	0.6458	0.6479	0.6482



(a) Silhouette of E1 and E2 by fold.



(b) Context-group ablation effects.

Figure 3. Supporting analyses of representation geometry and context contribution.**Table 14.** Zero-shot performance on 180 locked SHHS1 participants (seed 42). No weights were updated on SHHS. Intervals are subject-cluster bootstrap intervals for the subject-mean macro-F1. Transition macro-F1 is restricted to the 44,744 epochs within ± 1 of a stage change (26.5% of the sample).

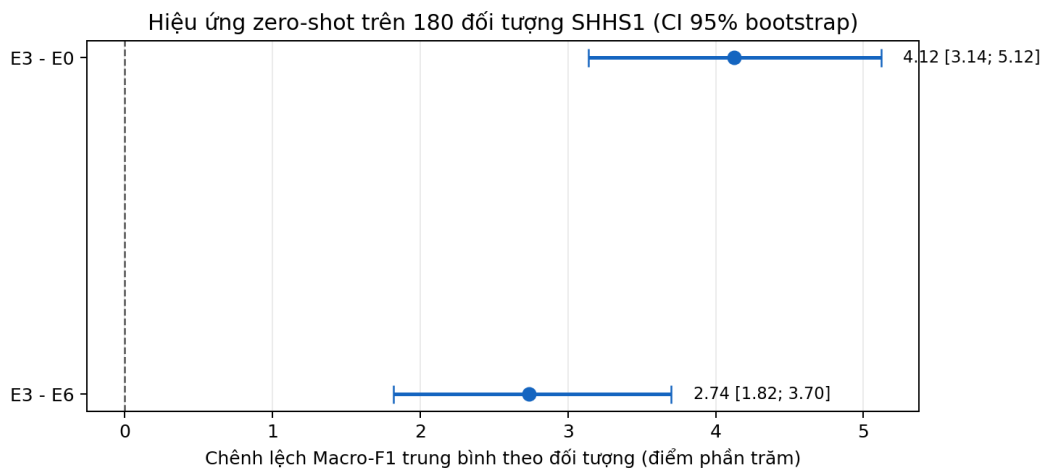
ID	Subject-mean macro-F1 (CI _{95%})	Pooled macro-F1	Accuracy	κ	F1 (N1)	Transition macro-F1
E0	0.5268 [0.5101, 0.5431]	0.5761	0.6648	0.5339	0.2998	0.4323
E1	0.5333	0.5822	0.6709	0.5409	0.3197	0.4352
E2	0.5205	0.5668	0.6656	0.5327	0.2833	0.4219
E3	0.5680 [0.5503, 0.5850]	0.6099	0.7016	0.5801	0.3451	0.4689
E6	0.5407 [0.5227, 0.5584]	0.5732	0.6742	0.5408	0.3102	—

Table 15. Seed-123 replication of the zero-shot campaign on the same 180 SHHS1 participants, extended to E4. Unlike the in-domain campaign, the ordering and the significance of both locked contrasts replicate.

ID	Subj. macro-F1	Pooled macro-F1	Acc.	κ	Contrast	Δ [CI _{95%}]	p_{Holm}
E0	0.5303	0.5785	0.6738	0.5420	E3 – E0	0.0331 [0.0225, 0.0438]	5.19×10^{-9}
E2	0.5409	0.5858	0.6874	0.5597	E3 – E6	0.0131 [0.0023, 0.0239]	0.0085
E3	0.5634	0.6031	0.7003	0.5772	E4 – E2	0.0323 [0.0236, 0.0410]	7.40×10^{-13}
E4	0.5732	0.6147	0.7064	0.5869	E3 – E4	–0.0098 [–0.0125, –0.0073]	1.89×10^{-10}
E6	0.5503	0.5865	0.6873	0.5552			

Table 16. Locked paired comparisons on SHHS1 subject-mean macro-F1 ($n = 180$). E3 – E2 and the seed-123 E4 rows are post-hoc or secondary and are excluded from the locked family.

Comparison	Role	Δ	CI _{95%}	p_{Holm}	W/T/L	Δ_{sign}
E3 – E0	locked primary	0.0412	[0.0314, 0.0512]	2.65×10^{-13}	138/0/42	+0.533
E3 – E6	locked primary	0.0274	[0.0182, 0.0370]	1.87×10^{-8}	125/0/55	+0.389
E1 – E0	secondary	0.00651	[0.00192, 0.01087]	0.00325	108/0/72	+0.200
E2 – E1	secondary	−0.01280	[−0.02209, −0.00350]	0.01005	80/0/100	−0.111
E3 – E2	post-hoc	0.04750	[0.03724, 0.05792]	2.35×10^{-17}	147/0/33	+0.633
E4 – E2 (seed 123)	post-hoc	0.03232	[0.02365, 0.04104]	7.40×10^{-13}	135/1/44	+0.508
E4 – E3 (seed 123)	post-hoc	0.00982	[0.00730, 0.01250]	1.89×10^{-10}	—	—

**Figure 4.** Subject-paired zero-shot effects on SHHS1.**Table 17.** Per-class transfer of E3 from Sleep-EDF to SHHS1, with the predicted-to-true epoch ratio as a marginal-calibration diagnostic. A ratio of 1.0 means the model emits the class at its true rate.

Stage	Sleep-EDF		SHHS1				$\Delta F1$
	F1	Pred/True	Prec.	Rec.	F1	Pred/True	
W	0.9304	0.993	0.8823	0.8014	0.8399	0.908	−0.0905
N1	0.5276	1.051	0.2528	0.5437	0.3451	2.151	−0.1824
N2	0.8514	0.984	0.7394	0.7349	0.7372	0.994	−0.1143
N3	0.8085	1.003	0.9440	0.2582	0.4055	0.273	−0.4030
REM	0.8343	1.017	0.6052	0.8939	0.7218	1.477	−0.1125

Table 18. Oracle decomposition of the E3 domain gap. Each row corrects one confusion channel in the SHHS1 confusion matrix and recomputes macro-F1. These are diagnostic counterfactuals, not achievable performance.

Condition	macro-F1	Gap recovered
E3 on SHHS1 (observed)	0.6099	—
+ N3→N2 errors corrected	0.7444	0.1345 (74.5%)
+ N2→REM errors corrected	0.6596	0.0497 (27.5%)
+ both corrected	0.7947	0.1848 (102.4%)
E3 on Sleep-EDF (reference)	0.7904	—

Table 19. Pooled confusion matrix of E3 on SHHS1 (rows: reference; columns: prediction). The two dominant off-diagonal cells are set in bold.

Reference \ Predicted	W	N1	N2	N3	REM
W	29,638	4,481	1,171	21	1,672
N1	874	3,807	601	3	1,717
N2	2,348	5,947	56,063	324	11,605
N3	78	39	16,674	5,888	127
REM	655	784	1,311	1	23,183

Table 20. Per-class metrics on SHHS1: E3 versus E2 (169,012 scored epochs).

Stage	E3 Prec.	E3 Rec.	E3 F1	E2 Prec.	E2 Rec.	E2 F1	$\Delta F1$
W	0.8823	0.8014	0.8399	0.8780	0.7275	0.7957	+0.0442
N1	0.2528	0.5437	0.3451	0.2367	0.3528	0.2833	+0.0618
N2	0.7394	0.7349	0.7372	0.7362	0.6937	0.7143	+0.0228
N3	0.9440	0.2582	0.4055	0.9659	0.2496	0.3967	+0.0087
REM	0.6052	0.8939	0.7218	0.4886	0.9450	0.6442	+0.0776

Table 21. Per-class metrics of E3 and E2 on SHHS1 restricted to the transition neighbourhood ($r = 1$; 44,744 epochs). Compare the N3 row with its whole-sample counterpart in Table ??.

Stage	Support	E3 Prec.	E3 Rec.	E3 F1	E2 Prec.	E2 Rec.	E2 F1
W	8,870	0.7501	0.5661	0.6452	0.7335	0.4749	0.5765
N1	5,654	0.3570	0.5644	0.4373	0.3615	0.3861	0.3734
N2	18,223	0.5739	0.6345	0.6027	0.5701	0.6195	0.5938
N3	7,563	0.8306	0.0733	0.1346	0.8607	0.0645	0.1200
REM	4,434	0.4026	0.7535	0.5248	0.3014	0.8561	0.4459
Overall (transition)		macro-F1 0.4689, acc. 0.5290, κ 0.3642			macro-F1 0.4219, acc. 0.4910, κ 0.3188		

5. Discussion

5.1. Principal findings

Within the primary campaign, E3 exceeded E6 by 0.0213 macro-F1 with a fully positive bootstrap interval and Holm significance, and the post-protocol repeat preserved a positive effect of 0.0102. Across two runs the evidence favours an amplitude-handling scheme that preserves relationships between recordings over one that normalises each recording independently, while the shifting Holm-adjusted p -value shows the inferential strength remains contingent on randomisation. The same ordering held out of domain.

The complete E3 pipeline outperformed the E0 control in both cohorts. On SHHS1 this was one of two comparisons locked before the test set was opened. Because E3 alters architecture and preprocessing together, the supportable conclusion is that E3 as an evaluated procedure transfers better within this protocol; credit for individual components must come from the component contrasts, and those tell a more complicated story.

5.2. Why the context ablation was null: receptive-field redundancy

The context ablation found no added value from the P and N feature groups (Table ??). Reported as a bare negative, this invites the wrong inference, namely that neighbouring epochs carry little information about stage transitions—a conclusion in tension with essentially the whole sequence-modelling literature [?, ?, ?].

The architecture resolves the tension. The C/P/N groups supply ± 1 epoch of context at the *encoder*. Downstream, the non-causal TCN operates over the entire overnight recording with a receptive field of 253 epochs, i.e. ± 126 epochs or roughly ± 63 min. The information that the P and N branches provide is thus a strict subset of what the sequence model already has direct access to, several times over. The ablation was not testing whether temporal context helps; it was testing whether supplying a ± 1 -epoch summary *twice* helps. A null result is the expected outcome, and the observed effect sizes— $|\Delta| < 0.0024$ on the transition criterion, intervals spanning zero—are exactly the size one would predict from redundant inputs.

This reframing has three consequences. It explains why the FULL condition can lose to CN (Table ??): redundant inputs add parameters and noise without adding information. It means the result licenses no conclusion whatsoever about the value of temporal context in sleep staging; it constrains only the marginal value of encoder-level context *given* a long-receptive-field sequence model. And it prescribes the design a genuine test would require: ablate the TCN’s receptive field itself—by restricting dilations, by making the network causal, or by windowing the input—rather than ablating features nested inside it. The sign-changing interaction index reported in Section ?? independently confirms that no scalar summary of the groups’ contribution is available from this design.

We note in passing that the descriptive asymmetry favouring the next epoch over the previous one (Section ??) survives this reframing as a hypothesis worth testing under a properly context-limited design, since prospective scoring rules give it an independent physiological rationale.

5.3. An amplitude-threshold account of the N3 collapse

Section ?? established that the transfer failure is localised in N3, has a high-precision/low-recall threshold signature, and runs counter to the label-prior shift. We propose the following account, which is consistent with every observation and makes falsifiable predictions.

Both scoring manuals define deep sleep partly through an absolute voltage criterion—slow waves exceeding roughly $75 \mu\text{V}$ peak-to-peak occupying a minimum fraction of the epoch [?, ?]. A model trained on Sleep-EDF learns this criterion implicitly, in the units and at the derivation it was trained on: Fpz–Cz, in a corpus of comparatively healthy subjects. Two things change in SHHS1. First, the derivation is C4–A1. The sleep slow oscillation has a pronounced frontal amplitude maximum and propagates antero-posteriorly [?], so identical slow-wave activity registers at substantially lower amplitude at a central-mastoid derivation than at a frontopolar one. Second, SHHS participants are markedly older, and slow-wave amplitude declines strongly with age [?]. Both effects push genuine N3 epochs below the learned amplitude criterion, in the same direction.

The account makes four predictions, and all four are borne out by data already collected. (i) N3 recall should collapse while N3 precision stays high or rises, because the epochs that still clear the threshold are

unambiguously N3—observed, precision 0.944 against recall 0.258. (ii) The misplaced epochs should go overwhelmingly to N2, the adjacent stage differing chiefly in slow-wave quantity—observed, 73.1% of all true N3. (iii) The effect should be insensitive to filtering, since filtering does not restore amplitude that the derivation and the ageing cortex never delivered—observed, N3 recall is 0.2582 for E3 and 0.2496 for E2 despite markedly different preprocessing, the only class where E3’s large preprocessing advantage essentially vanishes. (iv) The failure should be graded by proximity to the amplitude criterion, hence worst at the edges of N3 bouts where slow-wave activity is least established—observed, and strikingly: N3 recall falls further to 0.0733 in the transition neighbourhood (Table ??), with 90.6% of transition-region N3 going to N2, while precision there remains 0.831.

Prediction (iv) is the one that discriminates most sharply. A model that had simply failed to learn a transferable concept of N3 would degrade roughly uniformly across N3 epochs. A model applying a miscalibrated amplitude threshold must fail most where the true signal sits closest to that threshold. The observed three-and-a-half-fold gradient between whole-sample and transition-region recall is the signature of the latter.

There is an uncomfortable corollary. The design choice that makes E3 the best configuration overall—preserving absolute amplitude relationships rather than *z*-scoring each recording—is the same choice that ties the model’s N3 criterion to an absolute voltage scale and thus makes it montage-dependent. E3’s principal strength and its worst per-class failure share a cause. The evaluation is currently blind to this trade-off because macro-F1 averages the N3 collapse against gains elsewhere.

This yields a decisive and cheap experiment that we did not run and now recommend. Per-record *z*-scoring (E6) removes absolute scale and re-expresses amplitude relative to each recording’s own distribution, which is the quantity a montage change most nearly preserves. The account predicts that E6 should show *substantially better N3 recall on SHHS1 than E3*, even though E6 is worse overall (0.5407 vs 0.5680 subject-mean macro-F1). Per-class SHHS metrics for E6 were not computed in this campaign; obtaining them requires no retraining, only re-scoring existing predictions, and would either confirm the amplitude account or refute it outright. If confirmed, the practical implication is not to adopt E6 wholesale but to decouple the two roles—absolute scale for stages where it helps, relative scale for the N3 criterion—or to recalibrate the N3 decision threshold on a small target-domain sample. The 5 adaptation and 15 validation SHHS participants held out for exactly this purpose and never used remain available.

5.4. The roles of the TCN and the ResNet-1D encoder

In-domain, E1 – E0 and E2 – E1 gave positive descriptive gains that did not survive Holm correction. Out of domain, E1 – E0 stayed positive but small while E2 – E1 turned significantly negative. The architectural benefit is therefore domain-dependent, and the claim that ResNet-1D is uniformly superior to the control encoder is not supported—it is contradicted out of domain on unfiltered signals. The dependable advantage of ResNet-1D-TCN is operational: competitive accuracy with a much simpler pipeline and 3.76× faster inference, paid for with 4.37× the parameters and 28.4% more memory. As Section 4.4 notes, that speed-up comes from replacing 15 sequential CNN branches, not from the TCN.

The silhouette result is a useful negative: a widely used representation-quality index ranked the two encoders opposite to their downstream utility, in 10 folds out of 10. Per-epoch Euclidean cluster compactness is simply not the property a long-context sequence model needs, and we would caution against using such indices to select encoders for sequential tasks.

5.5. Preprocessing dominates transfer

The largest effect in the study is E3 – E2 on SHHS1 (+0.0475, $\Delta_{\text{sign}} = +0.633$), a contrast in which architecture is held fixed and only preprocessing changes. It is roughly seven times the largest in-domain architectural contrast. The E4 extension shows band-pass filtering alone captures most of this, with clipping and constant rescaling adding nothing for transfer. Since the largest architectural effect we measured out of domain was *negative*, the ordering is unambiguous within this study: preprocessing choices dominated architecture choices for cross-cohort generalisation. Given how much more attention architecture receives in this literature, we regard this as the finding with the widest implications, while noting that E3 changes

several operations at once and no single operation is identified as causal.

5.6. Positioning relative to published results

We deliberately do not present a table of accuracies from other papers alongside ours. The wake-trimming policy alone (Section 3.1) moves the wake fraction, and hence accuracy and macro-F1, by an amount comparable to every architectural effect measured here; subject-count, split scheme, and channel choice compound this. Published Sleep-EDF figures for the models discussed in Section 2 [?, ?, ?, ?] were obtained under protocols that differ in several of these dimensions simultaneously, so a numerical comparison would misrepresent the evidence in either direction. Our E0 control is a re-implementation of the ZleepAnlystNet design evaluated under *our* protocol, and it is the appropriate reference point for every number in this paper. Readers wishing to situate these results externally should compare protocols first, and we have specified ours completely for that purpose.

5.7. Limitations and a prioritised agenda

Results should be read within the data and protocol evaluated. SHHS1 is one specific out-of-domain sample, not a population. The second seed corroborates direction but two fixed seeds cannot characterise initialisation variance; a proper treatment would require enough seeds to estimate a distribution, and we report no such claim. Several component analyses were run after the sample was opened and are supporting evidence only. Single-channel EEG limits separability for N1 and REM in principle, and the evaluation window uses ground-truth anchors, so it is an offline benchmarking device rather than a deployable rule. We also report no calibration analysis: the sequence models are trained with unweighted cross-entropy and their probability estimates are untested, which matters for any clinical use where an abstain option would be valuable [?, ?].

Two reporting gaps should be stated plainly. Absolute-performance intervals are available for the SHHS configurations (Table ??) but not for the Sleep-EDF ones, where only difference intervals were retained. And although paired win/tie/loss counts and median subject-level differences were preserved for every comparison, the full per-subject macro-F1 distributions were not archived, so we cannot report the dispersion (IQR, range) of individual performance or identify which participants the models handle worst. For a metric whose clinical relevance is inherently per-patient this is a real limitation, and we would archive those vectors in any future campaign. Finally, per-class SHHS metrics exist only for E2 and E3; obtaining them for E0, E1 and E6 requires no retraining and would settle several questions raised above, including the decisive one in Section 5.3.

Ordered by expected value, we would prioritise: (1) computing per-class SHHS metrics for E6 to test the amplitude account, which requires no retraining; (2) recalibrating the N3 threshold on the held-out SHHS adaptation participants, since 74.5% of the domain gap sits in that single channel; (3) ablating the TCN receptive field directly, which is the experiment the context ablation was unable to perform; (4) estimating initialisation variance with a sufficient number of seeds; and (5) reporting calibration and selective-prediction behaviour. Items (1) and (2) are inexpensive and address the largest single source of error identified in this study.

6. Conclusion

A unified subject-wise design with a single locked test set allowed six configurations to be compared fairly on Sleep-EDF. In the primary campaign E3 attained a descriptive macro-F1 of 0.7904 and exceeded E6 with a fully positive interval and Holm significance; a post-protocol repeat preserved the direction of that effect. Isolated substitutions of the sequence model and the epoch encoder produced descriptive or domain-dependent gains, but no stable advantage survived multiple-comparison correction. ResNet-1D-TCN simplifies the pipeline and is 3.76× faster at inference, at the cost of more parameters and memory.

On SHHS1, E3 outperformed both locked references. The largest effect anywhere in the study, however, was a preprocessing contrast under fixed architecture, and the largest architectural effect out of domain was negative. Within this protocol, preprocessing—not architecture—governed cross-cohort transfer.

The two analyses we regard as most consequential are corrective rather than confirmatory. The cross-cohort gap is not diffuse degradation: two confusion channels account for essentially all of it, and the dominant one

is a slow-wave amplitude-threshold failure that is diagnosable from its precision–recall signature, that runs opposite to the label-prior shift, and that is plausibly correctable without retraining. And the null result of the context ablation was structurally predetermined by a sequence model whose receptive field already subsumed the context under test, so it constrains experimental design rather than physiology. The contribution of this work is therefore a protocol with out-of-domain evidence, a quantified operational trade-off, and a demonstration that careful decomposition of where a model fails is more informative than the aggregate score that summarises it.

Reproducibility and data availability

Source code, experimental protocols, data partitions, intermediate results and summary tables are retained in the project repository. Sleep-EDF Expanded is publicly available from PhysioNet. SHHS is accessed through the NSRR under its data use agreement; derived data and participant identifiers are not published. All primary results are cross-checked against an evidence register and accompanying manifest files, allowing traceability from data to analysis. Every checkpoint is verified by SHA-256 before loading, and source provenance is recorded for each result-affecting module change.

Ethics

This study is a secondary analysis of de-identified data from research repositories with their own governance mechanisms; no new participants were recruited. Use of SHHS follows NSRR access conditions. All interpretations apply to the samples and protocols described and do not constitute clinical validation.

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Declaration of competing interests

The authors declare no competing interests.